

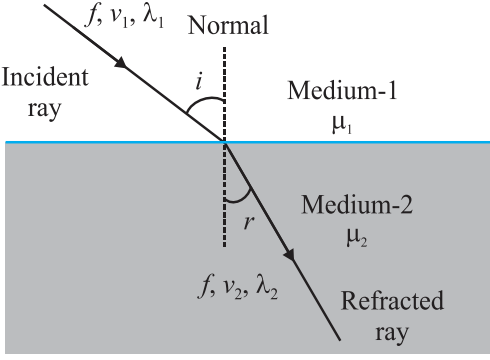
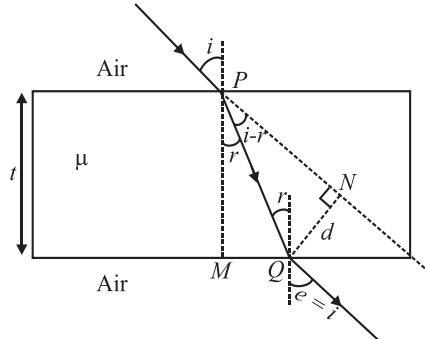


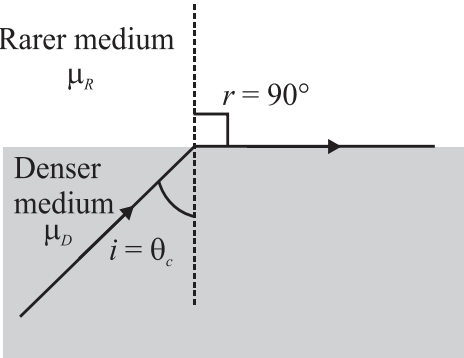
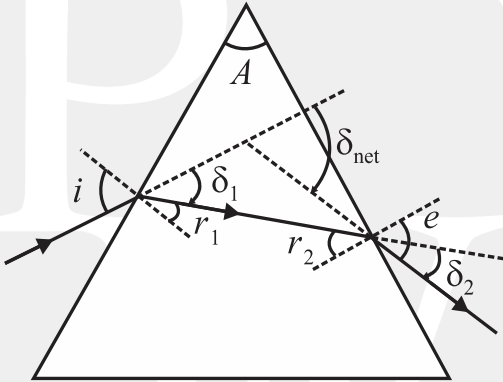
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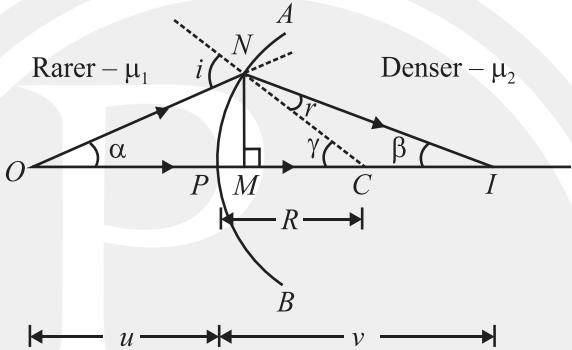
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Ray Optics and Optical Instruments

Sr. No.	Concept	Formulas	Other information
1.	Formulas related to spherical mirror	For any spherical mirror, $f = \frac{R}{2}$ Mirror formula, $\frac{1}{u} + \frac{1}{v} = \frac{1}{f} = \frac{2}{R}$ Magnification, $m = \frac{h_2}{h_1} = -\frac{v}{u} = \frac{f}{f-u} = \frac{f-v}{f}$	f = focal length. R = radius of curvature. u = object distance from pole of the mirror. v = image distance from pole of the mirror. h_2 = image height. h_1 = object height.
2.	Refractive index	Refractive index: $\mu = \frac{c}{v}$ Relative refractive index: ${}_1\mu_2 = \mu_{21} = \frac{\mu_2}{\mu_1}$ ${}_2\mu_1 = \mu_{12} = \frac{\mu_1}{\mu_2}$ $\Rightarrow \mu_{21} = \frac{1}{\mu_{12}} \Rightarrow \mu_{21} \times \mu_{12} = 1$	c = speed of light in a vacuum. v = speed of light in a medium. μ_2 = refractive index of 2nd medium. μ_1 = refractive index of 1st medium. μ_{21} = Refractive index of medium-2 with respect to medium-1 μ_{12} = Refractive index of medium-1 with respect to medium-2
3.	Cauchy's formula	According to Cauchy's formula, $\mu = A + \frac{B}{\lambda^2}$	A, B = constants. λ = wavelength. μ = refractive index of medium.

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4.	Snell's Law	 $\frac{\sin i}{\sin r} = \text{constant (Snell's Law)}$ $\frac{\sin i}{\sin r} = \mu_{21} \Rightarrow \frac{\sin i}{\sin r} = \frac{\mu_2}{\mu_1}$ $\mu_1 \sin i = \mu_2 \sin r$	i = the angle of incidence. r = the angle of refraction. f = frequency. v_2 = velocity of light in 2 nd medium. v_1 = velocity of light in 1 st medium.
5.	Another form of snell's law	From Snell's law $\frac{\sin i}{\sin r} = \frac{\mu_2}{\mu_1} = \frac{\lambda_1}{\lambda_2} = \frac{v_1}{v_2}$	λ_2 = wavelength of light in 2 nd medium. λ_1 = wavelength of light in 1 st medium.
6.	Apparent depth and change of speed due to refraction	$\frac{\text{Apparent depth}}{\text{Actual depth}} = \frac{d'}{d} = \mu$ Apparent depth of object at the bottom of a vessel filled with two immiscible liquids of refractive indices μ_1 and μ_2 $= \frac{d_1}{\mu_1} + \frac{d_2}{\mu_2}$ Normal shift in the position of the object $x = d \left(1 - \frac{1}{\mu} \right)$	v_{IS} = velocity of image. v_{OS} = velocity of object. d_1 and d_2 = depth of two liquids respectively. d = real depth
7.	Shift due to refraction through glass slab	Lateral Shift: $d = \frac{t}{\cos r} \sin(i - r)$ 	t = thickness of the slab i = the angle of incidence. r = the angle of refraction.

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8.	Critical Angle (θ_c)	 $\sin \theta_c = \frac{\mu_R}{\mu_D}$ If rarer medium is air: $\sin \theta_c = \frac{1}{\mu}$	μ_D = refractive index of denser medium. μ_R = refractive index of rarer medium.
9.	Refraction through prism	 Angle of prism: $A = r_1 + r_2$ Angle of deviation: $\delta_{net} = (i + e) - A$ Condition for minimum deviation: $i = e$, $r_1 = r_2 = \frac{A}{2}$ Relation between minimum deviation and refractive index: From, $\delta = i + e - A$ $\delta_m = 2i - A \Rightarrow i = \frac{\delta_m + A}{2}$ $\frac{\mu_p}{\mu_s} = \frac{\sin\left(\frac{\delta_m + A}{2}\right)}{\sin\left(\frac{A}{2}\right)}$	A = angle of prism. e = angle of emergence. r_1, r_2 = angle of refraction on two surfaces. μ_s = refractive index of the surroundings. μ_p = refractive index of the prism.

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10.	Dispersion of light related formula	Deviation produced by a prism of small angle, $\delta = (\mu - 1)A$ Angular dispersion = $\delta_V - \delta_R = (\mu_V - \mu_R)A$ Dispersive power, $\omega = \frac{\delta_V - \delta_R}{\delta} = \frac{\mu_V - \mu_R}{\mu - 1}$ Mean deviation, $\delta = \frac{\delta_V + \delta_R}{2}$ Mean refractive index, $\mu = \frac{\mu_V + \mu_R}{2}$	μ_R = refractive index due to red light. μ_V = refractive index due to violet light. δ_R = deviation of red light. δ_V = deviation of violet light.
11.	Refraction through a spherical surface	 For refraction from rarer to denser medium, $\frac{\mu_2}{v} - \frac{\mu_1}{u} = \frac{\mu_2 - \mu_1}{R}$ Refraction from denser to rarer medium; $\frac{\mu_1}{v} - \frac{\mu_2}{u} = \frac{\mu_1 - \mu_2}{R}$	R = radius of curvature of spherical surface. μ_2 = refractive index of denser medium. μ_1 = refractive index of rarer medium.
12.	Lens maker's formula	For the lens placed in air, $\frac{1}{f} = (\mu_g - 1) \left[\frac{1}{R_1} - \frac{1}{R_2} \right]$ Focal length of lens immersed in liquid: $\frac{1}{f_l} = (\mu_g - 1) \left(\frac{1}{R_1} - \frac{1}{R_2} \right) = \left(\frac{\mu_g}{\mu_l} - 1 \right) \left(\frac{1}{R_1} - \frac{1}{R_2} \right)$	${}^a\mu_g$ = refractive index of glass w.r.t. air R_1, R_2 = the radii of curvature of its spherical surfaces. f = focal length of lens. ${}^l\mu_g$ = refractive index of glass w.r.t liquid ${}^a\mu_l$ = refractive index of liquid w.r.t air

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13.	Formulas related to lens	For a thin lens: $\frac{1}{v} - \frac{1}{u} = \frac{1}{f}$ Magnification = $\frac{\text{size of image}}{\text{size of object}}$ $m = \frac{h_2}{h_1} = \frac{v}{u} = \frac{f}{f+u} = \frac{f-v}{f}$	u = object distance. v = image distance. f = focal length of lens.
14.	Power of lens	Mathematically, $P = \frac{1}{f(m)}$ For lens $P = \frac{1}{f} = (\mu - 1) \left[\frac{1}{R_1} - \frac{1}{R_2} \right]$	P = power of lens
15.	Lens combinations	Magnifications produced by the combination of lenses: $m = m_1 \times m_2 \times m_3 \times \dots$ The combined focal length of two lenses f: $\frac{1}{f} = \frac{1}{f_1} + \frac{1}{f_2}$ For n thin lenses in contact, $\frac{1}{f} = \frac{1}{f_1} + \frac{1}{f_2} + \dots + \frac{1}{f_n}$ or $P = P_1 + P_2 + \dots + P_n$ When the two thin lenses are separated by a distance d, $\frac{1}{f} = \frac{1}{f_1} + \frac{1}{f_2} + \frac{d}{f_1 f_2}$ or Power, $P = P_1 + P_2 + (d \times P_1 \times P_2)$	f = focal length of the combination. P = power of the lens combination. m = magnification of the combination. d = separation between two lenses.
16.	Rayleigh's law of scattering	The intensity of light wave: $I \propto \frac{1}{\lambda^4}$	λ = wavelength of light.
17.	Simple microscope.	When the final image is formed at the least distance of distinct vision, $m = 1 + \frac{D}{f}$ When the final image is formed at infinity, $m = \frac{D}{f}$	m = magnifying power D = least distance of distinct vision. f = focal length of lens.

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18.	Compound microscope.	Magnifying power, $m = m_o \times m_e$ When the final image is formed at the least distance of distinct vision, $m = \frac{-v_o}{u_o} \left(1 + \frac{D}{f_e} \right) = -\frac{L}{f_o} \left(1 + \frac{D}{f_e} \right)$ When the final image is formed at infinity, $m = \frac{-v_o}{u_o} \cdot \frac{D}{f_e} = -\frac{L}{f_o} \cdot \frac{D}{f_e}$	L = the length of the tube f_o = focal length of the objective lens. f_e = focal length of the eyepiece. m_o = magnification of the objective lens. m_e = magnification of the eyepiece. v_o = image distance from the objective lens. u_o = object distance from the objective lens.
19.	Astronomical telescopes	(i) In normal adjustment, $m = \frac{f_o}{f_e}$ Distance between objective and eyepiece $\Rightarrow L = f_o + f_e$ (ii) When the final image is formed at the least distance of distinct vision, $m = \frac{-f_o}{f_e} \left(1 + \frac{f_e}{D} \right)$ Distance between objective and eyepiece $\Rightarrow L = f_o + \frac{f_e D}{f_e + D}$	m = magnification of the telescope. D = least distance of distinct vision. L = length of telescope